

Rational Numbers

Số hữu tỉ – Theory & Worked Examples (Lý thuyết & Ví dụ mẫu)

In this lesson you will learn to / Trong bài này em sẽ học:

- Recognise and write rational numbers
- Opposite and reciprocal numbers
- Compare and order rational numbers
- Add, subtract, multiply, divide
- Powers of rational numbers
- Order of operations
- Solve simple equations
- Read maths aloud in English

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1 What is a rational number? (Số hữu tỉ là gì?)

Definition / Định nghĩa

A **rational number** is a number that can be written as a fraction $\frac{a}{b}$ with $a, b \in \mathbb{Z}$ and $b \neq 0$. The set of all rational numbers is written $\mathbb{Q}$.

Số hữu tỉ là số viết được dưới dạng phân số $\frac{a}{b}$ với a, b là số nguyên, $b \neq 0$. Tập hợp các số hữu tỉ kí hiệu là $\mathbb{Q}$.

Read: a over b , where a and b are integers and b is not equal to zero

Which numbers are rational? Every integer, fraction, finite decimal and mixed number is rational:

$$5 = \frac{5}{1}, \quad -3 = \frac{-3}{1}, \quad 0,6 = \frac{6}{10} = \frac{3}{5}, \quad 1\frac{1}{4} = \frac{5}{4}, \quad -\frac{3}{4}.$$

So $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q}$. (Số tự nhiên $\subset$ số nguyên $\subset$ số hữu tỉ.)

1.1 Equivalent fractions (Phân số bằng nhau)

Rule / Quy tắc

Multiply or divide the numerator and the denominator by the *same* non-zero number:

$$\frac{a}{b} = \frac{a \cdot m}{b \cdot m} = \frac{a : n}{b : n} \quad (m, n \neq 0, n \text{ a common factor}).$$

Example / Ví dụ

Complete: $\frac{3}{4} = \frac{?}{20}$. Since $4 \times 5 = 20$, multiply the top by 5 too: $\frac{3 \times 5}{4 \times 5} = \frac{15}{20}$.

1.2 The number line (Trục số)

Positive numbers are on the right of 0, negative numbers on the left. **The farther right, the greater.**



1.3 Opposite and reciprocal (Số đối và số nghịch đảo)

Opposite number (số đối)

The opposite of x is $-x$. Their sum is 0:

$$x + (-x) = 0.$$

$$\text{opp}\left(\frac{2}{5}\right) = -\frac{2}{5}, \quad \text{opp}\left(-\frac{7}{3}\right) = \frac{7}{3}.$$

Reciprocal (số nghịch đảo)

If $x \neq 0$, the reciprocal of x is $\frac{1}{x}$. Their product is 1:

$$\frac{a}{b} \cdot \frac{b}{a} = 1 \quad (a \neq 0).$$

reciprocal of $\frac{3}{4}$ is $\frac{4}{3}$; of -5 is $-\frac{1}{5}$.

Common mistake / Lỗi hay gặp

0 **has no reciprocal** (we cannot divide by 0). So the statement “every rational number has a reciprocal” is *false*; it is true for every rational number *except* 0.

Số 0 không có số nghịch đảo.

2 Comparing and ordering (So sánh và sắp xếp)

Rule / Quy tắc

Three ways to compare rational numbers:

- 1. Common denominator** (*quy đồng mẫu số*): make denominators equal (and positive), then compare numerators.
- 2. Decimals** (*đổi ra số thập phân*): write all numbers as decimals.
- 3. Number line** (*trục số*): the number on the right is greater.

Also: *negative* $< 0 < \text{positive}$; and for two negatives, the one with the larger absolute value is smaller.

Example / Ví dụ

Compare $-\frac{5}{6}$ and $-\frac{3}{4}$.

Common denominator 12: $-\frac{5}{6} = -\frac{10}{12}$, $-\frac{3}{4} = -\frac{9}{12}$. Since $-10 < -9$, we get

$$-\frac{5}{6} < -\frac{3}{4}.$$

Read: minus five over six is less than minus three over four

Example / Ví dụ

Put in **increasing order** (*thứ tự tăng dần*): $-\frac{1}{2}$, 0,4, $-0,75$, $\frac{2}{3}$.

Decimals: $-0,5$; 0,4; $-0,75$; 0,666... Order: $-0,75 < -0,5 < 0,4 < 0,666...$

$$-\frac{3}{4} < -\frac{1}{2} < \frac{2}{5} < \frac{2}{3}.$$

Remember / Ghi nhớ

Increasing = small $\rightarrow$ large (*tăng dần*). **Decreasing** = large $\rightarrow$ small (*giảm dần*).

2.1 A rational number between two others (Số hữu tỉ nằm giữa)

Example / Ví dụ

Find x with $-\frac{2}{3} < x < -\frac{1}{2}$.

Use denominator 12: $-\frac{8}{12} < x < -\frac{6}{12}$, so $x = -\frac{7}{12}$ works. For more answers use denominator 24:

$$-\frac{16}{24} < x < -\frac{12}{24}, \text{ so } x \in \left\{-\frac{13}{24}, -\frac{14}{24}, -\frac{15}{24}\right\}.$$

3 Adding and subtracting (Cộng và trừ)

Rule / Quy tắc

Write both fractions with a **common denominator**, then add/subtract the numerators:

$$\frac{a}{b} \pm \frac{c}{b} = \frac{a \pm c}{b}, \quad x - y = x + (-y).$$

Example / Ví dụ

$$(a) \frac{3}{4} - \frac{5}{6} = \frac{9}{12} - \frac{10}{12} = \frac{-1}{12} = -\frac{1}{12}.$$

$$(b) -\frac{2}{5} + \frac{7}{10} = -\frac{4}{10} + \frac{7}{10} = \frac{3}{10}.$$

$$(c) \frac{5}{6} - \left(-\frac{1}{4}\right) = \frac{5}{6} + \frac{1}{4} = \frac{10}{12} + \frac{3}{12} = \frac{13}{12}.$$

Read: five over six minus open parenthesis minus one over four close parenthesis

Common mistake / Lỗi hay gặp

Learner writes $-\frac{2}{3} + \frac{5}{6} = -\frac{4}{6} - \frac{5}{6}$. **Wrong!** The sign of the second fraction must not change:

$$-\frac{2}{3} + \frac{5}{6} = -\frac{4}{6} + \frac{5}{6} = \frac{1}{6}.$$

Common mistake / Lỗi hay gặp

$\frac{1}{2} - \frac{3}{4} \neq \frac{1-3}{2-4}$. Never subtract the numerators and the denominators separately. Correct: $\frac{2}{4} - \frac{3}{4} = -\frac{1}{4}$.

4 Multiplying and dividing (Nhân và chia)

Rule / Quy tắc

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{a \cdot c}{b \cdot d}, \quad \frac{a}{b} : \frac{c}{d} = \frac{a}{b} \cdot \frac{d}{c} \quad (c \neq 0).$$

To divide, **multiply by the reciprocal of the divisor** (nhân với số nghịch đảo của số chia).

Signs: same signs $\Rightarrow$ positive; different signs $\Rightarrow$ negative.

Example / Ví dụ

$$(a) -\frac{4}{9} \cdot \frac{15}{8} = -\frac{4 \cdot 15}{9 \cdot 8} = -\frac{60}{72} = -\frac{5}{6} \quad (\text{tip: cancel first, } \frac{4}{8} = \frac{1}{2} \text{ and } \frac{15}{9} = \frac{5}{3}, \text{ so } -\frac{1 \cdot 5}{2 \cdot 3} = -\frac{5}{6}).$$

$$(b) -\frac{3}{8} : \frac{9}{16} = -\frac{3}{8} \cdot \frac{16}{9} = -\frac{48}{72} = -\frac{2}{3}.$$

$$(c) \frac{3}{4} : (-6) = \frac{3}{4} \cdot \left(-\frac{1}{6}\right) = -\frac{3}{24} = -\frac{1}{8}.$$

Read: minus three over eight divided by nine over sixteen equals minus two over three

Remember / Ghi nhớ

Distributive law (tính chất phân phối) saves time:

$$\frac{3}{7} \cdot \frac{11}{5} + \frac{3}{7} \cdot \frac{4}{5} = \frac{3}{7} \left(\frac{11}{5} + \frac{4}{5} \right) = \frac{3}{7} \cdot 3 = \frac{9}{7}.$$

5 Powers of rational numbers (Lũy thừa)

Definition / Định nghĩa

For $n \in \mathbb{N}, n \geq 1$: $\left(\frac{a}{b}\right)^n = \underbrace{\frac{a}{b} \cdot \frac{a}{b} \cdots \frac{a}{b}}_{n \text{ factors}} = \frac{a^n}{b^n}$.

Here $\frac{a}{b}$ is the **base** (cơ số) and n is the **exponent** (số mũ).

Read: open parenthesis a over b close parenthesis raised to the n -th power

Rule / Quy tắc

For any rational x and $m, n \in \mathbb{N}$:

Product (nhân)

$$x^m \cdot x^n = x^{m+n}$$

Quotient (chia)

$$x^m : x^n = x^{m-n} \quad (x \neq 0, m \geq n)$$

Power of a power (lũy thừa của lũy thừa)

$$(x^m)^n = x^{m \cdot n}$$

Example / Ví dụ

(a) $\left(-\frac{2}{3}\right)^3 = \frac{(-2)^3}{3^3} = -\frac{8}{27}$ (negative base, odd exponent $\Rightarrow$ negative).

(b) $\left(-\frac{1}{2}\right)^4 = \frac{1}{16}$ (even exponent $\Rightarrow$ positive).

(c) $\left(\frac{2}{5}\right)^5 : \left(\frac{2}{5}\right)^3 = \left(\frac{2}{5}\right)^2 = \frac{4}{25}$.

(d) $\left(\frac{3}{4}\right)^2 \cdot \left(\frac{3}{4}\right)^3 = \left(\frac{3}{4}\right)^5$.

Read: open parenthesis minus one over two close parenthesis raised to the fourth power equals one over sixteen

Common mistake / Lỗi hay gặp

$\left(-\frac{1}{2}\right)^2 = \frac{1}{4}$ but $-\frac{1}{2}^2$ would mean $-\frac{1}{4}$. Always use brackets around a negative base!

6 Order of operations (Thứ tự thực hiện phép tính)

Rule / Quy tắc

Brackets $\rightarrow$ **Powers** $\rightarrow$ **$\times$ and $:$** $\rightarrow$ **$+$ and $-$**

Ngoặc $\rightarrow$ lũy thừa $\rightarrow$ nhân/chia $\rightarrow$ cộng/trừ. Cùng cấp thì làm từ trái sang phải; ngoặc theo thứ tự $() \rightarrow [] \rightarrow \{\}$.

Example / Ví dụ

Calculate $1 - \left[\frac{2}{5} - \left(-\frac{3}{10}\right)\right] \cdot \frac{1}{2}$.

Step 1 (inner brackets): $\frac{2}{5} + \frac{3}{10} = \frac{4}{10} + \frac{3}{10} = \frac{7}{10}$.

Step 2 (multiply): $\frac{7}{10} \cdot \frac{1}{2} = \frac{7}{20}$.
 Step 3 (subtract): $1 - \frac{7}{20} = \frac{20}{20} - \frac{7}{20} = \frac{13}{20}$.

Example / Ví dụ

Calculate $\left(-\frac{1}{3}\right)^2 + \frac{5}{6} : \left(-\frac{5}{2}\right)$.
 Power first: $\frac{1}{9}$. Division: $\frac{5}{6} \cdot \left(-\frac{2}{5}\right) = -\frac{10}{30} = -\frac{1}{3}$. Then $\frac{1}{9} - \frac{1}{3} = \frac{1}{9} - \frac{3}{9} = -\frac{2}{9}$.

7 Solving simple equations (Giải phương trình đơn giản)

Rule / Quy tắc

Do the **inverse operation** (*phép tính ngược*) on both sides:

$$x + a = b \Rightarrow x = b - a; \quad x - a = b \Rightarrow x = b + a; \quad a \cdot x = b \Rightarrow x = b : a \quad (a \neq 0).$$

Example / Ví dụ

(a) $x - \frac{2}{3} = \frac{1}{4} \Rightarrow x = \frac{1}{4} + \frac{2}{3} = \frac{3}{12} + \frac{8}{12} = \frac{11}{12}$.
 (b) $\frac{2}{5}x = -\frac{4}{15} \Rightarrow x = -\frac{4}{15} : \frac{2}{5} = -\frac{4}{15} \cdot \frac{5}{2} = -\frac{20}{30} = -\frac{2}{3}$.
 (c) $x + \frac{1}{2} = \frac{5}{6} \Rightarrow x = \frac{5}{6} - \frac{1}{2} = \frac{5}{6} - \frac{3}{6} = \frac{2}{6} = \frac{1}{3}$.

Remember / Ghi nhớ

Check your answer (*thử lại*): put x back into the equation. For (c): $\frac{1}{3} + \frac{1}{2} = \frac{2}{6} + \frac{3}{6} = \frac{5}{6} \checkmark$

8 Real-life problems (Bài toán thực tế)

Example / Ví dụ

A tank is $\frac{2}{3}$ full. Then $\frac{1}{5}$ of the **total** capacity is used. What fraction is still full?
 $\frac{2}{3} - \frac{1}{5} = \frac{10}{15} - \frac{3}{15} = \frac{7}{15}$ Answer: $\frac{7}{15}$ of the tank.

Example / Ví dụ

The temperature at 6 a.m. is $-2,5^{\circ}\text{C}$. It rises by $6,4^{\circ}\text{C}$, then falls by $3,1^{\circ}\text{C}$.
 $-2,5 + 6,4 - 3,1 = 3,9 - 3,1 = 0,8$. Answer: $0,8^{\circ}\text{C}$.

Remember / Ghi nhớ

Mixed units? Convert everything to **one form** (all fractions or all decimals) before calculating.

9 How to read maths aloud (Cách đọc bằng tiếng Anh)

Expression	Read aloud
$-\frac{3}{4}$	minus three over four
$\frac{7}{8}$	seven over eight
$\frac{2}{5} \times \frac{3}{4} = \frac{3}{10}$	two over five times three over four equals three over ten
$\left(-\frac{1}{2}\right)^4$	open parenthesis minus one over two close parenthesis raised to the fourth power
3^5	three raised to the fifth power
$-\frac{2}{3} < -\frac{1}{2} < 0$	minus two over three is less than minus one over two; minus one over two is less than zero
$x + \frac{1}{2} = \frac{3}{4}$	x plus one over two equals three over four

10 Vocabulary (Từ vựng)

English	IPA	Vietnamese	Example
rational number	/ˈræʃənəl ˈnʌmbə(r)/	số hữu tỉ	$-\frac{3}{4}$
number line	/ˈnʌmbə(r) laɪn/	trục số	$-\frac{1}{2} < 0 < \frac{1}{2}$
fraction	/ˈfrækʃən/	phân số	$\frac{7}{8}$
numerator	/ˈnjuːməreɪtə(r)/	tử số	$\frac{5}{8}$: numerator is 5
denominator	/dɪˈnɒmɪneɪtə(r)/	mẫu số	$\frac{5}{8}$: denominator is 8
common denominator	/ˈkɒmən dɪˈnɒmɪneɪtə(r)/	mẫu số chung	$\frac{1}{2} = \frac{3}{6}, \frac{1}{3} = \frac{2}{6}$
opposite number	/ˈɒpəzɪt ˈnʌmbə(r)/	số đối	$\frac{2}{5} + (-\frac{2}{5}) = 0$
reciprocal	/rɪˈsɪprəkəl/	số nghịch đảo	$\frac{2}{3} \times \frac{3}{2} = 1$
power	/ˈpaʊə(r)/	lũy thừa	$(-\frac{1}{2})^4$
exponent	/ɪkˈspəʊnənt/	số mũ	3^5 : exponent is 5
equivalent fraction	/ɪˈkwɪvələnt ˈfrækʃən/	phân số bằng nhau	$\frac{1}{2} = \frac{2}{4}$
order of operations	/ˈɔːdə(r) ˌɒpəˈreɪʃənz/	thứ tự thực hiện phép tính	$2 + 3 \times 4 = 14$
increasing order	/ɪnˈkriːsɪŋ ˈɔːdə(r)/	thứ tự tăng dần	$-1 < 0 < \frac{1}{2}$
decreasing order	/dɪˈkriːsɪŋ ˈɔːdə(r)/	thứ tự giảm dần	$\frac{3}{4} > 0 > -\frac{1}{2}$
calculate	/ˈkælkjuleɪt/	tính toán	$\frac{5}{8} - \frac{1}{4} = \frac{3}{8}$
equation	/ɪˈkweɪʒən/	phương trình	$x + \frac{1}{2} = \frac{3}{4}$

11 Quick review (*Tóm tắt*)

Rational numbers at a glance

- ▶ $\mathbb{Q} = \left\{ \frac{a}{b} : a, b \in \mathbb{Z}, b \neq 0 \right\}; \mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q}.$
- ▶ Opposite of x is $-x$ ($x + (-x) = 0$); reciprocal of $\frac{a}{b}$ is $\frac{b}{a}$ ($a \neq 0$).
- ▶ Compare: common denominator $\rightarrow$ decimals $\rightarrow$ number line. Farther right = greater.
- ▶ Add/subtract: common denominator. Divide: multiply by the reciprocal.
- ▶ $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}; x^m x^n = x^{m+n}; x^m : x^n = x^{m-n}; (x^m)^n = x^{mn}.$
- ▶ Order: brackets $\rightarrow$ powers $\rightarrow \times / : \rightarrow + / -.$
- ▶ Equations: use the inverse operation, then check.